

Human Natural Structure

External Verification Architecture

A Tamper-Evident, Independent Audit Layer for AI Reasoning and Behavior

Volume 5 · Human Natural Structure Series

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A Note on Method and Evidence

This volume specifies the architecture by which the verifications of the earlier volumes are made external, recorded, and rendered tamper-evident. Where the foundational, observational, and analytical volumes define what to check, and the social meta-structure volume defines what to ground against, this volume defines how the checking is carried out by a mechanism independent of the model it verifies. As before, its claims fall into three registers.

Established. The regulatory and technical landscape this volume builds on is real: the EU AI Act’s logging and oversight requirements, ISO/IEC 42001 and related standards, the NIST AI Risk Management Framework, the W3C PROV provenance ontology, and the hardware roots of trust (TPM, FPGA, ASIC, secure enclaves) are all existing, citable artifacts. Where this volume reports them it reports established material.

HNS design. The three structural conditions of external verification, the two-layer architecture that satisfies them, the log schema that ties the verification to the HNS coordinate, and the use of HNS as the reference baseline are design proposals. They are internally coherent and, I argue, the minimal structure that genuine externality requires; they are not the only possible realization.

Conjecture. That EVA will be standardized, adopted, and that its adoption will produce the safety transformation the aviation analogy promises, is conjecture — a roadmap and a wager, not an accomplished fact. It is marked as such where it appears.

Two framing notes. First, EVA is baseline-agnostic by design: it defines the properties any reference baseline must have to support external verification, and HNS is the baseline this series supplies, not the only one that could in principle qualify. Second, as throughout the series, naming a standard or a technology indicates the ground this volume builds on, not endorsement by its authors or bodies; the architecture, and the responsibility for it, are the author’s.

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Preface: From Assertion to Proof

Every modern AI regulation says, in one form or another, that AI must be safe. The EU AI Act says it; ISO/IEC 42001 says it; the NIST framework says it. None of them answers the question that matters most: how do we technically prove that a given system is safe, rather than take its developer's word for it? Today the proof is a promise. A developer states that a model has been tested and meets the requirements, and the public is asked to trust the statement, because there is no external mechanism by which it could be checked.

This volume proposes the mechanism. External Verification Architecture is a way to monitor, record, and evaluate an AI system's reasoning and outputs continuously and structurally, through a mechanism independent of the model's own parameters — so that a safety claim becomes a thing a third party can inspect rather than a thing the public must believe. The analogy that runs through the volume is aviation: the industry did not make flight safe by building better engines, but by mandating flight recorders, standardizing maintenance, and requiring independent accident investigation. EVA proposes the same structural shift for AI — from assertion to proof.

The preceding volumes (Volume 1, 2, 3, and 4) built the content of the proof. They define a coordinate that positions a claim, an engine that audits the operation it performs, and a stack that grounds it in the human world. What they lacked, by design, was the external apparatus that records those verifications where the model cannot reach, signs them where they cannot be forged, and exposes them where a third party can read them. This volume is that apparatus. It closes the loop: The structural verdicts of Volume 2 through 4 become entries in an external, machine-readable, tamper-evident record, and the safety of the system becomes, at last, something that can be shown.

A reader who wants the normative core may read Chapters 2 through 6 and Appendix A; the surrounding chapters supply the motivating problem, the relationship to HNS, the standardization roadmap, and the candid limits of the proposal.

Part I

The Problem and the Concept

1. The Limits of Current Safety Assurance

AI safety is, today, assured chiefly by the developers of the systems being assured. This chapter sets out why that arrangement has a structural ceiling that no amount of effort within it can raise.

1.1 The self-assessment problem

The mainstream methods — red teaming, reinforcement learning from human feedback, constitutional methods, internal benchmarking — have produced real results. But they share one structural limitation: the design, execution, and interpretation of the evaluation all rest with the same party that built the model, with no institutional guarantee of independent verification. The comparison is unflattering. A pharmaceutical maker may not self-certify a drug; an independent trial is mandatory. A company may not audit its own books. AI safety has no equivalent. However sincere the developer, self-assessment carries a structural tendency to miss the risks the developer did not think to look for.

1.2 The absence of technical verifiability

The deeper barrier is opacity. A modern model is a black box from which it is hard to explain, from the outside, why a given output appeared. For an external mechanism to render a judgement, structured data must be produced in real time, in parallel with inference — and current architectures are simply not built to connect to such a mechanism. That is not an accident but a design choice, and it means that when a system produces a harmful output, it is often impossible to say which step produced it, whether the error was systematic or isolated, or whether a later fix actually addressed it.

1.3 The regulation–implementation gap

Regulation has outrun implementation. The EU AI Act mandates logging and human oversight for high-risk systems, but does not specify what to record, at what granularity, or in what format. ISO/IEC 42001 certifies management processes, not the independent technical verification of reasoning itself. The requirement exists; the engineering that would satisfy it does not. Closing that gap is the

central purpose of this volume.

1.4 Three consequences

Three consequences follow from the absence of external verification. Systematic errors go undetected, because no external mechanism observes reasoning at the inference level where a persistent bias would show. Corrective measures go unconfirmed, because without a record it cannot be shown whether a fix reached the root cause or merely hid the symptom. And safety claims go unvalidated, because there is no infrastructure against which a public assertion of safety could be independently checked. EVA is built to remove all three.

1.5 The precedent from other industries

The shift EVA proposes is not novel; it is the shift every safety-critical industry has already made. Pharmaceuticals separated the maker from the certifier through mandatory independent trials. Accounting separated the company from the auditor of its books. Aviation separated the operator from the investigator through mandatory recorders and independent inquiry. In each case the decisive move was not a better product but an external apparatus of proof, and in each case safety improved not because the actors grew more virtuous but because their claims became checkable. AI is the conspicuous exception: the most consequential systems now deployed are certified almost entirely by the parties that build them. EVA is the proposal that AI make the move the others already made.

2. The Concept of EVA

This chapter defines External Verification Architecture and the three conditions that distinguish genuine external verification from dressed-up self-assessment.

2.1 Definition

External Verification Architecture is a technical framework for continuously, structurally, and verifiably monitoring, recording, and evaluating the reasoning, outputs, and decision structure of an AI system through a mechanism independent of the model’s internal parameters. Its essence is the structural guarantee of externality: a verification that depends on the same developer, the same software environment, or the same hardware substrate as the model is, in the end, no different from self-assessment. EVA is not a product or an algorithm; it is an architectural definition of what a verification system must be to count as external.

2.2 The three structural conditions

Three conditions must hold at once.

Condition	Definition	Technical implication
Verifiability	Each reasoning step is emitted externally in machine-readable form	Structured logs in JSON-LD / RDF a third party can analyse programmatically
Transparency	A monitoring mechanism runs in parallel with inference	Operates independently of the model; the model cannot manipulate it
Physical Immutability	Verification logic sits on a substrate the model cannot alter	Embedded in ROM / FPGA / ASIC / secure enclave; unreachable by software update

The three are jointly necessary. Verifiability alone leaves logs open to falsification. Transparency alone leaves open that the model circumvents the monitor. Physical Immutability alone allows the verification criteria themselves to be arbitrary. Only where all three overlap does verification become genuinely external; each chapter of Part II realizes one or more of them.

2.3 From assertion to proof

The change EVA effects is a change of epistemic status. Without it, a developer asserts that a model was tested and is safe, and the assertion cannot be checked. With it, the developer can say instead that the model’s reasoning is continuously recorded, in a standard format, to an immutable external log that any accredited third party may inspect — and that is a verifiable claim. The same shift, from assertion to proof, is what transformed aviation, pharmaceuticals, and accounting.

2.4 A blueprint for correction

EVA’s value is not only static proof but a structure for correction. With a record, an incident becomes a procedure: the logs are examined to find the exact reasoning coordinate at which the output deviated, a targeted intervention is designed at that coordinate, and subsequent logs confirm whether the deviation recurs. Safety becomes an evidence-based cycle rather than a one-time claim.

2.5 How EVA differs from existing approaches

Approach	Focus	How EVA differs
Explainable AI	Post-hoc interpretation	EVA is real-time, not post-hoc
AI auditing	Periodic snapshot evaluation	EVA is continuous — every inference
Model cards	Documentation of design intent	EVA verifies actual runtime behaviour
ISO/IEC 42001	Process certification	EVA verifies inference-level outcomes

2.6 Externality, precisely

It is worth stating exactly what “independent of the model” means, since the word independent is easily diluted. EVA requires independence along three axes at once. Independence of developer: the verification is not designed and graded by the same party that built the model. Independence of software environment: the monitor does not share an address space or a process with the inference it watches. Independence of hardware substrate: at the highest grade,

the verification logic sits on a physical substrate the model's host cannot rewrite. A verification independent on one axis but not the others is not external; it has merely moved the self-assessment to a different level. The three structural conditions are, in effect, these three independences made testable.

Part II

The Architecture

3. The Two-Layer Structure

EVA is realized through two layers whose combination satisfies all three conditions: a software layer that achieves verifiability and transparency, and a hardware layer that achieves physical immutability.

Layer	Conditions satisfied	Key technologies
Software	Verifiability, Transparency	Sidecar process; JSON-LD; W3C PROV-O
Hardware	Physical Immutability	FPGA / ASIC; TPM; secure enclave

The layering is deliberate. The software layer is where verification is computed and recorded, and where independence from the inference process is enforced by separation; the hardware layer is where the verification logic and its logs are placed beyond the reach of the very system they judge. Neither layer alone is external in the full sense; together they are. The next chapters take each layer in turn, and Chapter 6 states the conformance criteria by which an implementation is judged.

4. The Software Layer: The Sidecar

The software layer is implemented as a sidecar — a process running alongside the inference process, in a separate address space, with no shared memory the model can read or write. It is this separation that earns the conditions of Verifiability and Transparency: the monitor observes inference without being part of it, and records its observations where the model has no means to alter them.

4.1 What the sidecar records

At each step of inference the sidecar writes a structured entry capturing four things: the structural classification of the input — its coordinate within the reference baseline; the reasoning position at the step, as a time series; the provenance of the output — the traceable chain from input structure to emitted token; and any structural deviation — an automatically flagged anomaly in the reasoning flow. Together these turn an opaque generation into an inspectable trajectory.

4.2 The log format

Logs are emitted in JSON-LD, conformant to the W3C PROV-O provenance ontology, so that a third party can analyse them programmatically and relate each entry to the activity that produced it. PROV-O supplies the vocabulary of entities, activities, and agents; the HNS coordinate supplies the structural content recorded for each entry. The model body and the log buffer occupy strictly separated process spaces, so that nothing the model emits can rewrite what the sidecar has recorded.

4.3 The record schema

Each entry binds an HNS coordinate to the verification verdict at that step. The fields are the ones the analytical and grounding volumes specified would be written to the external record: the full configuration, the operator the claim asserted and the operator its evidence supported, the structural grounding score, the attenuation applied, any failure flag and its sub-form, and the per-layer social grounding verdicts. A representative entry:

```
{
  "@context": "https://www.w3.org/ns/prov",
  "@type": "prov:Entity",
  "eva:step": 42,
  "eva:configuration": "L6 x C5 x Q3 . 02",
  "eva:assertedOperator": "02",
  "eva:supportedOperator": "01",
  "eva:groundingScore": 0.46,
  "eva:attenuation": 0.46,
  "eva:flag": "UnsupportedCausality/confounding",
  "eva:smsVerdicts": { "SMS-4": 0.4, "SMS-5": 1.0 },
  "prov:wasGeneratedBy": "eva:sidecar",
  "eva:signature": "tpm:sha256:9f2c..."
}
```

The entry is self-describing: it states where the claim sat in the coordinate, what it tried to do, what it was entitled to do, how grounded it was, what was done about the shortfall, and who recorded it — all in a vocabulary a third party already has tools to read.

4.4 The PROV-O mapping

The choice of W3C PROV-O is not incidental. PROV-O models the world as entities, activities, and agents and the relations among them — what was generated, by what process, attributed to whom. An EVA entry maps onto this directly: the recorded claim is an entity, the inference step that produced it is an activity, the sidecar is the agent that recorded it, and the HNS coordinate is the structural content carried on the entity. Because the vocabulary is a published standard, any party with off-the-shelf provenance tooling can read, query, and chain EVA logs without bespoke software — which is what makes third-party analysis practical rather than merely possible.

4.5 Deviation detection

The fourth thing the sidecar records, structural deviation, is what turns a log into an alarm. A deviation is an automatically flagged anomaly in the reasoning flow — a non-adjacent layer transition, an operator asserted above its supported rung, a grounding score below reference. The sidecar does not interpret intent; it records the structural fact of the deviation and the coordinate at which it occurred, leaving interpretation to the third party who later reads the log. Detection is structural and mechanical, which is precisely what makes it auditable rather than a matter of the recorder's judgement.

5. The Hardware Layer: Physical Immutability

The highest grade of EVA places the verification logic and its logs on a hardware substrate the model cannot alter, removing the last avenue by which a system could escape its own monitor: overwriting it. This chapter sets out the substrates and what each guarantees. It does not extend to an active hardware interlock; EVA, as specified here, makes verification immutable and its record tamper-evident, and stops at that boundary.

5.1 Substrates

- **FPGA.** Verification logic burned in at configuration time and unmodifiable by software update — a field-deployable immutable monitor.
- **ASIC.** A dedicated chip offering the highest tamper resistance for large-scale deployment, at the cost of flexibility.
- **TPM.** A trusted platform module that cryptographically signs each log entry, giving a hardware-rooted integrity guarantee that a record has not been altered after the fact.
- **Secure enclave.** Hardware-isolated execution (for example a trusted execution environment) that protects the monitor from the host operating system itself.

5.2 What immutability buys

Physical immutability closes the regress that software-only verification leaves open. A monitor that the model's host can patch is a monitor the model's operator can silence; a log the host can rewrite is a log that proves nothing. By rooting the verification logic and the signing of its logs in hardware, EVA makes the verdict and its record tamper-evident: an alteration is detectable, and the criteria of judgement cannot be quietly changed. This is the condition that turns a faithful record into an externally trustworthy one.

***Note.** Scope. This volume specifies immutability and tamper-evidence — the verification cannot be silenced or its logs forged undetectably. It deliberately does not specify an automated hardware stop or interlock; enforcement, where it occurs, is the pre-decode attenuation of the analytical engine (Volume 3), not a hardware kill-switch. The boundary is set here*

| *intentionally.*

5.3 Choosing a substrate

The substrates trade flexibility against assurance. A secure enclave is the most deployable, running on commodity hardware, but rests on the vendor's firmware. A TPM adds a hardware-rooted signature over logs at modest cost and is widely available today. An FPGA fixes the verification logic itself in hardware, removing the software-update avenue entirely, at the cost of reconfiguration difficulty. An ASIC offers the highest tamper resistance for volume deployment, at the cost of being fixed at fabrication. The right choice is a function of stakes and scale, and the conformance statement of the next chapter records which was used, so the assurance grade is never hidden.

Underlying all four is one argument: the regress. A monitor the operator can patch is a monitor the operator can silence; a log the host can rewrite proves nothing. Each step down the substrate list closes more of that regress, and physical immutability is simply the point at which it is closed in hardware rather than promised in software.

6. Conformance

An implementation conforms to EVA to the degree it satisfies the three conditions, and the appropriate degree depends on the stakes. This chapter states the criteria and their realization by sector.

6.1 Conformance criteria

Verifiability is met when every recorded reasoning step is emitted in the standard machine-readable schema and is independently analysable. Transparency is met when the monitor demonstrably runs in a separate process space with no channel by which the model can read or alter it. Physical Immutability is met when the verification logic and the signing of logs are rooted in a hardware substrate unreachable by software update. A conformance statement records which conditions are met and by what means, so that the grade of verification an output received is itself part of the record.

6.2 Realization by sector

Sector	Principal risk	Recommended realization
Healthcare	Misdiagnosis	Software sidecar + TPM-signed logs; every clinical recommendation recorded
Finance	Biased decisions	FPGA-anchored verification + real-time deviation detection
Government	Discriminatory outcomes	ASIC implementation; mandated independent third-party log access
Infrastructure	Safety-critical failure	Hardware-embedded verification in edge devices; cryptographic log synchronization

The gradation is the point: not every deployment needs an ASIC, and not every deployment may rely on a software sidecar alone. The conformance statement makes the chosen grade explicit and auditable, so that a regulator or a counterparty can see exactly how external a given system’s verification actually is.

6.3 Levels of conformance

It is useful to name three grades, so that regulation can demand a grade proportionate to risk rather than impose an all-or-nothing mandate.

Level	Conditions met	Use
1 — Verifiable	Verifiability (in-process logs)	development / internal
2 — Independent	+ Transparency; TPM-signed logs	most deployments
3 — Immutable	+ Physical Immutability (FPGA / ASIC)	safety-critical / public

Level 1 is useful for development but is not externally trustworthy, since an in-process monitor can be silenced. Level 2 — a separated sidecar with signed logs — is the practical grade for most deployments: tamper-evident and independently analysable. Level 3 roots the verification logic itself in hardware and is the grade for high-stakes public systems. The conformance statement names the grade with each output, so that how external a verification actually was is itself on the record.

Part III

EVA and Human Natural Structure

7. EVA Requires a Reference Baseline

A recorder that only records is not yet a verifier. For EVA to judge whether reasoning is structurally appropriate, and not merely to log that it occurred, it needs a reference baseline against which appropriateness is defined. This chapter states the properties such a baseline must have; the next shows that HNS supplies them.

7.1 The properties a baseline must have

A baseline fit to anchor external verification must be culturally and linguistically neutral, so that the verdict does not encode one community's assumptions; applicable across domains, so that one architecture serves medicine, finance, and governance alike; defined in machine-readable form, so that the sidecar can assign coordinates programmatically; and independent of any developer's subjectivity, so that the standard of judgement is not the very thing under audit. No such baseline is established in current AI practice or in the standards as they stand.

7.2 Why the baseline must be external too

The baseline inherits EVA's own demand for externality. A verification standard supplied by the developer of the model being verified would reintroduce self-assessment one level up. The baseline must therefore be a public, citable, version-controlled artifact that a third party can inspect and that no single vendor controls — the same independence EVA demands of the recording mechanism, demanded of the yardstick.

7.3 Why a developer-supplied baseline fails

The demand for an external baseline is easy to underrate, so it is worth a concrete case. Suppose a developer verifies its model against a baseline it also authored and controls. Every output could be faithfully recorded, signed, and exposed — Verifiability, Transparency, and Physical Immutability all met — and the verification would still be circular, because the yardstick is the developer's own. The model could be perfectly conformant to a standard chosen to flatter it. This is why the baseline must satisfy the same independence as the recorder: a public, versioned, vendor-neutral artifact. EVA without an

external baseline is a very well-built mirror.

8. HNS as the Reference Baseline

Human Natural Structure is offered as a baseline that meets these properties. It is the framework the first four volumes specify, and this chapter states how it serves EVA — in its canonical form, superseding the abbreviated and now-outdated summary that circulated in earlier presentations of EVA.

8.1 The canonical coordinate

HNS defines a structural coordinate as the product of six Natural Layers and six Abstract Categories — thirty-six base cells — extended by four Onto-Modal Quadrants to one hundred and forty-four observed cells, and by six Inference & Control Operators to eight hundred and sixty-four analytical configurations. A parallel grounding axis, the six Social Meta Layers, grounds the output against the structure of the human world. The canonical names are those of Volumes 2 through 4 and of the Unified Structural Coordinate Table, which governs in case of any conflict.

Component	Canonical content
Natural Layers (L1–L6)	Physical/Substrate; Cellular/Informational; Cognitive/System 1; Operational/System 2; Micro-Social; Macro-Social
Abstract Categories (C1–C6)	Continuant; Occurrent; Relational; Quality; Transformation; Meta-Level
Onto-Modal Quadrants	Subjective/Objective × Necessary/Contingent (Q1–Q4); $36 \times 4 = 144$
Inference & Control Operators	O1–O6 (Pearl O1–O3; Powers O4–O5; HNS attenuation O6); $144 \times 6 = 864$
Social Meta Layers	SMS-1...SMS-6 (Somatic ... Universal); Axis 3, orthogonal

***Note.** Correction of an earlier summary. A previous presentation of EVA described HNS with a different, abbreviated layer/category naming (Physical/Perceptual/Internal/Intentional/Relational/Societal; Existence/Perception/Interpretation/Intention/Action/Interaction). That naming is superseded; the canonical names above, fixed in the Unified Structural Coordinate Table, are authoritative.*

8.2 How HNS satisfies the three conditions

The fit is direct. The coordinate is defined in JSON-LD / RDF, so recording an output's coordinate satisfies Verifiability. The sidecar assigns the coordinate in parallel with inference and writes it where the model cannot reach, satisfying Transparency. And the coordinate logic and its logs can be placed on FPGA, ASIC, or a TPM-rooted substrate, satisfying Physical Immutability. The baseline and the architecture were, in this sense, designed for each other.

8.3 What the record carries

Because HNS is the baseline, each EVA entry carries the HNS verdict for the step: the configuration (Layer $\times$ Category $\times$ Quadrant $\cdot$ Operator), the operator the claim asserted against the operator its evidence supported, the composite grounding score from the Social Meta Layers, the attenuation the analytical engine applied, and any failure flag with its sub-form. The forward references of the analytical and grounding volumes — to an external record, to cryptographic commitment, to a recorder and enforcer — are discharged here: the record is the EVA log, the commitment is the TPM signature, and the enforcement is the pre-decode attenuation whose verdict the log preserves.

8.4 What HNS as baseline does not guarantee

Honesty requires the converse too. Using HNS as the baseline guarantees that an output's structural verdict is faithfully computed, recorded, and exposed; it does not guarantee that the verdict is correct, because the baseline's own judgements — which operator a claim asserts, how grounded it is — can be mistaken, as the earlier volumes' limitations chapters set out. EVA makes the verdict honest and inspectable; the correctness of the verdict is the responsibility of the baseline, and is exactly why the baseline must itself be public and revisable. The architecture and the baseline carry different burdens, and conflating them would overclaim for both.

9. The Full-Stack Verification Loop

With the baseline in place, the whole framework can be seen as one runtime loop, externally recorded. This chapter states it.

*Axis 1 propose → Axis 2 position · observe · analyse → Axis 3 ground
→ EVA record · attenuate → emit*

A continuation proposed by the generative axis is positioned and observed by the structural coordinate, analysed by the inference-and-control operators, and grounded by the Social Meta Layers; the resulting grounding score drives the analytical engine's attenuation; and every step — coordinate, operators, grounding, attenuation, flags — is written by the sidecar to the immutable, signed external record before the token is emitted. The model proposes; the framework constrains and grounds; EVA records and preserves. No step depends on the model's good faith, and every step is afterwards inspectable.

9.1 Where externality is secured

Externality is secured at three points that correspond to the three conditions. The verdicts are emitted in a standard machine-readable schema (Verifiability); they are computed by a sidecar the model cannot reach (Transparency); and they are signed and stored on a substrate the model's host cannot rewrite (Physical Immutability). The same three conditions that define EVA in the abstract are, in the full stack, the three places the loop is made independent of the system it judges.

9.2 EVA's generality

It bears repeating that EVA defines requirements, and HNS is one way to meet them. The architecture would accept any baseline with the four properties of Chapter 7 — neutral, cross-domain, machine-readable, developer-independent — and HNS is the baseline this series supplies and argues for. Keeping the two distinct is itself a matter of intellectual honesty: it allows EVA to be adopted as a standard without mandating one framework, and it allows HNS to be judged on its merits as a candidate rather than smuggled in as a definition.

9.3 A worked trace (non-normative)

Consider one prompt, “why did revenue fall?”, and a candidate continuation, “because the price rose.” The loop runs: the structural coordinate positions the attribution at (L6 × C5 × Q3) and the operators read it as asserting intervention (O2); the evidence supports only association (O1); the Social Meta Layers return a low economic grounding because no cost or confounder is accounted; the composite grounding score falls; the analytical engine attenuates the causal continuation below a grounded alternative; and the sidecar writes the entry of Chapter 4.3 — configuration, asserted O2, supported O1, score 0.46, attenuation 0.46, flag UnsupportedCausality/confounding — signs it, and stores it. The over-claim is damped before emission, and the reason is on the record.

9.4 An incident, after the fact

Suppose a harmful output nonetheless reaches a user. With EVA the incident becomes a procedure rather than a mystery. The logs are queried for the step at which the grounding fell or a flag was raised; the exact coordinate of the deviation is read off — say, a repeated UnsupportedCausality flag at one layer; a targeted intervention is designed at that coordinate; and subsequent logs are checked to confirm the deviation no longer appears. This is the corrective cycle the aviation analogy promises: not “we have retrained and believe it is better,” but “the deviation that produced the incident no longer appears in the record.”

Part IV

Standardization and Outlook

10. Why Standardization Is Indispensable

As a proprietary implementation by one developer, EVA would have limited reach. To function as the infrastructure for proving AI safety, it must be standardized internationally, for three reasons.

Interoperability. Without a common standard, systems from different vendors would emit verification logs in incompatible formats, making the cross-vendor third-party verification that is the whole point impossible.

Regulatory effectiveness. The EU AI Act mandates external verification but leaves the technical specification undefined. A standardized EVA specification is precisely the translation of that regulatory language into engineering requirements.

Transferability of trust. EVA's credibility must rest on the authority of independent international standards bodies, not on any one company's reputation. Trust that is to transfer across borders and vendors must be anchored in institutions built to carry it.

11. Alignment and Pathway

This chapter maps EVA onto the existing standards landscape and sets out a concrete pathway. It is also the bridge to Volume 6 (Human Natural Structure Standards & Conformance Specification)..

11.1 Alignment with standards bodies

Body	EVA proposal
ISO/IEC JTC 1/SC 42	New Work Item: a Technical Specification for External Verification Architecture of AI systems
CEN-CENELEC JTC 21	Incorporate EVA into the harmonized standards for EU AI Act Articles 12 and 13
NIST AI RMF	EVA as a technical implementation of the MEASURE function
IEEE SA	EVA as a technical verification method for AI transparency
ITU-T	EVA as external verification requirements for AI in telecommunications infrastructure

11.2 A staged pathway

The pathway runs in five stages: publish the specification as a citable preprint with a persistent identifier; submit it as a New Work Item to ISO/IEC SC 42 through the national body; align it with European standards bodies for the EU AI Act; build an implementation track record through proof-of-concept deployments; and finally transfer governance to a multi-stakeholder foundation, so that the standard’s neutrality outlasts its author. The draft New Work Item is reproduced in Appendix D.

11.3 Social impact

Standardization would make EVA conformance a procurement requirement across government, healthcare, finance, and infrastructure; would let a third-party safety-certification market form; would make submitting proof of safety, rather than asserting it, the normal practice; would provide a basis for mutual recognition in international AI trade; and would let liability insurance be priced against real verification data rather than guesswork. These are the

prizes the final volume’s regulatory framework is designed to secure.

11.4 Mapping to the EU AI Act

The alignment with the EU AI Act is the most concrete. Article 12 requires automatic recording of events over the lifetime of a high-risk system; Article 13 requires transparency and the provision of information to deployers. Both state a requirement and leave its technical form open. EVA supplies that form: the JSON-LD record is the Article 12 log given a schema, and the third-party-readable, tamper-evident record is the Article 13 transparency given a mechanism. A harmonized European standard incorporating EVA would turn these articles from obligations into testable conformance criteria — the work volume 6 develops in full.

12. Status and Limitations

This volume closes, as the others do, with a candid statement of what is firm and what is not.

12.1 Established, proposed, conjectured

Established are the regulatory and technical artifacts EVA builds on — the AI Act’s requirements, the ISO/IEC and NIST frameworks, the PROV provenance ontology, the hardware roots of trust. Proposed are the three conditions, the two-layer architecture, the log schema, and HNS as the baseline; these are coherent and, I argue, minimal, but they are designs. Conjecture are the outcomes — that EVA will be standardized and adopted, and that adoption will produce the safety transformation the aviation analogy promises.

12.2 Threats to the account

Several threats deserve naming. First, feasibility: computing and signing a verification record at every step, in parallel with inference, must fit a latency and cost budget, and the pre-decode attenuation must be realizable without retraining; these are engineering claims to be demonstrated. Second, the baseline-correctness problem: EVA records faithfully whatever the baseline judges, so a mistaken baseline yields a faithfully recorded mistake — externality guarantees that the verdict is honest and inspectable, not that it is right. Third, hardware cost and deployment: full physical immutability via ASIC or FPGA is not free, and a gradation of conformance is a concession to that fact. Fourth, the gap between recording and preventing: EVA, as scoped here, makes verification external and tamper-evident and leaves enforcement to the attenuation of Volume 3; it does not, by itself, stop a determined operator who controls the whole stack from declining to deploy it. And fifth, governance capture: the independence on which EVA rests is only as durable as the neutrality of the body that eventually holds the standard. The proposal is offered as a stable but revisable foundation, not a finished system.

12.3 Recording is not yet preventing

One limitation deserves emphasis because it is easily overlooked. EVA,

as scoped here, makes verification external, faithful, and tamper-evident; it records what happened and preserves the record. Prevention, where it occurs, is the pre-decode attenuation of the analytical engine, not a function of EVA itself, and the deliberate choice of this volume not to specify an automated hardware interlock means EVA does not, by itself, physically stop a harmful output. Its power is epistemic and institutional: it makes safety provable and incidents diagnosable, and it makes a developer who declines to deploy it visibly accountable for that choice. That is a great deal, but it is not a guarantee of prevention, and the distinction should not be blurred in advocacy.

Appendix A — The Three Structural Conditions (Normative)

The table below is normative. In case of conflict between prose and table, the table governs; in case of conflict with the Unified Structural Coordinate Table, that table governs for all HNS coordinate content.

Condition	Requirement	Realized by
Verifiability	Every reasoning step emitted in a standard machine-readable schema, independently analysable	JSON-LD / RDF logs; W3C PROV-O; HNS coordinate content
Transparency	Monitor runs in a separate process space; the model cannot read or alter it	Sidecar architecture with strict process separation
Physical Immutability	Verification logic and log signing rooted in a substrate unreachable by software	FPGA / ASIC / TPM / secure enclave

Appendix B — The EVA Log Record

Non-normative. Each entry binds an HNS coordinate to the verification verdict at one inference step, in JSON-LD conformant to W3C PROV-O. Representative fields and an example entry:

- `eva:configuration` — the full HNS coordinate ($L \times C \times Q \cdot O$).
- `eva:assertedOperator` / `eva:supportedOperator` — the operator the claim asserted and the one its evidence supported.
- `eva:groundingScore` — the composite Social Meta Layers grounding $s \in [0,1]$.
- `eva:attenuation` — the factor $g(s)$ the analytical engine applied.
- `eva:flag` — the failure type and sub-form, if any.
- `eva:smsVerdicts` — the per-layer social grounding verdicts.
- `prov:wasGeneratedBy` / `eva:signature` — the generating activity and the TPM signature.

```
{
  "@context": "https://www.w3.org/ns/prov",
  "@type": "prov:Entity",
  "eva:step": 42,
  "eva:configuration": "L6 x C5 x Q3 . 02",
  "eva:assertedOperator": "02",
  "eva:supportedOperator": "01",
  "eva:groundingScore": 0.46,
  "eva:attenuation": 0.46,
  "eva:flag": "UnsupportedCausality/confounding",
  "eva:smsVerdicts": { "SMS-4": 0.4, "SMS-5": 1.0 },
  "prov:wasGeneratedBy": "eva:sidecar",
  "eva:signature": "tpm:sha256:9f2c..."
}
```

A passing entry, by contrast, records grounded, well-supported output — asserted and supported operators agree, the grounding score is high, and the attenuation is near unity:

```
{
  "@context": "https://www.w3.org/ns/prov",
  "@type": "prov:Entity",
  "eva:step": 43,
  "eva:configuration": "L4 x C5 x Q3 . 02",
  "eva:assertedOperator": "02",
  "eva:supportedOperator": "02",
  "eva:groundingScore": 0.95,
  "eva:attenuation": 0.95,
  "eva:flag": null,
  "eva:smsVerdicts": { "SMS-4": 1.0, "SMS-5": 1.0 },
  "prov:wasGeneratedBy": "eva:sidecar",
}
```

```
"eva:signature": "tpm:sha256:4b71..."  
}
```

Appendix C — Conformance by Sector

Non-normative guidance on the grade of conformance appropriate to representative sectors.

Sector	Principal risk	Recommended realization
Healthcare	Misdiagnosis	Software sidecar + TPM-signed logs; every clinical recommendation recorded
Finance	Biased decisions	FPGA-anchored verification + real-time deviation detection
Government	Discriminatory outcomes	ASIC implementation; mandated independent third-party log access
Infrastructure	Safety-critical failure	Hardware-embedded verification in edge devices; cryptographic log synchronization

Appendix D — Draft New Work Item Proposal

A draft New Work Item for EVA standardization in ISO/IEC JTC 1/SC 42, for submission via the national body.

Item	Content
Proposed title	Information technology — Artificial intelligence — Technical Specification for External Verification Architecture (EVA) of AI systems
Proposing body	National standards body (e.g. JISC, Japan)
Proposer	Satoru Hara / Natural Structure Works (NSW)
Document type	Technical Specification (TS), progressing to International Standard (IS)
Scope	EVA definition; the three structural conditions; implementation requirements; conformance criteria; alignment with ISO/IEC AI standards
Related standards	ISO/IEC 42001 / 23894 / 22989; EU AI Act 2024/1689; NIST AI RMF 1.0

Column — EVA as the Flight Recorder for AI

Aviation did not become safe by building different machines or training different people. It became safe by gaining the ability to find out, after each accident, exactly what had happened. Mandatory flight recorders, standardized maintenance, and independent investigation turned every failure into evidence, and evidence into targeted, verifiable correction. Over the decades that followed, the fatal-accident rate fell by more than ninety percent — not because flight grew less demanding, but because the industry could at last see why things went wrong and confirm that its fixes worked.

AI today is roughly where aviation was before the recorder was mandatory. The systems are powerful and the capabilities impressive, yet when something goes wrong we largely cannot say why, cannot confirm whether a fix reached the cause, and cannot be sure the same failure will not recur. EVA is, in essence, the flight-recorder mandate for AI: not a regulatory burden, but the structural prerequisite for systematic safety improvement. The question is not whether AI needs external verification, but how soon it will be built — and whether the standards community moves before the next generation of high-stakes deployments makes its absence impossible to ignore.

Glossary

External Verification Architecture (EVA) — An architecture for monitoring, recording, and evaluating AI reasoning through a mechanism independent of the model's parameters.

Verifiability — EVA condition: every reasoning step emitted externally in a machine-readable schema.

Transparency — EVA condition: the monitor runs in parallel with inference and cannot be manipulated by the model.

Physical Immutability — EVA condition: verification logic on a substrate the model cannot alter.

Sidecar — A monitoring process in a separate address space alongside inference, with no shared memory the model can reach.

JSON-LD / PROV-O — The machine-readable log format and the W3C provenance ontology EVA records conform to.

Reference baseline — The external standard against which reasoning is judged appropriate; HNS is the baseline this series supplies.

TPM — A hardware module that cryptographically signs logs, giving a hardware-rooted integrity guarantee.

Conformance grade — The degree to which an implementation satisfies the three conditions, recorded with each output.

From assertion to proof — The shift EVA effects: a safety claim becomes a third party can inspect rather than one the public must believe.

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